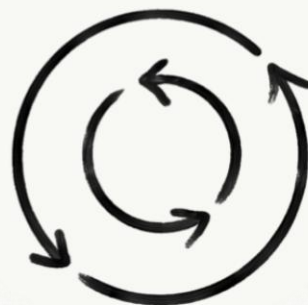


REPEAT



Prompt Repetition Improves Non-Reasoning LLMs

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Transformer Attention Limitation

- Most LLMs are trained as causal language models:
 - Tokens can attend to past tokens.
 - Tokens cannot attend to future tokens.
- **order dependencies** in prompts.

<CONTEXT> <QUESTION>  <QUESTION> <CONTEXT>

- The paper proposes:

<Query> <Query>

Method	Template	Example Query
Baseline	<QUERY>	Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>.
Prompt Repetition	<QUERY> <QUERY>	Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>. Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>.

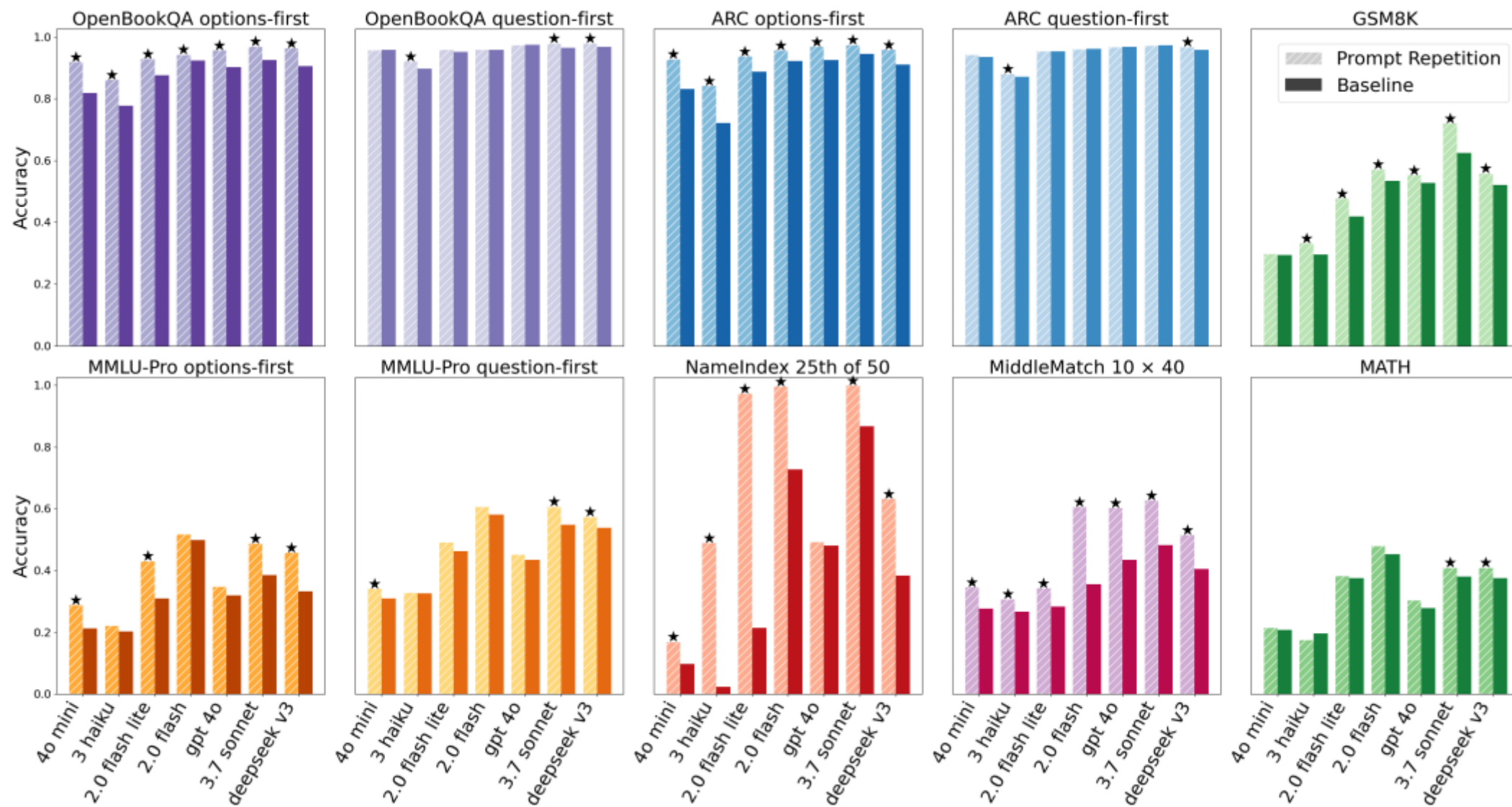
Experimental Results

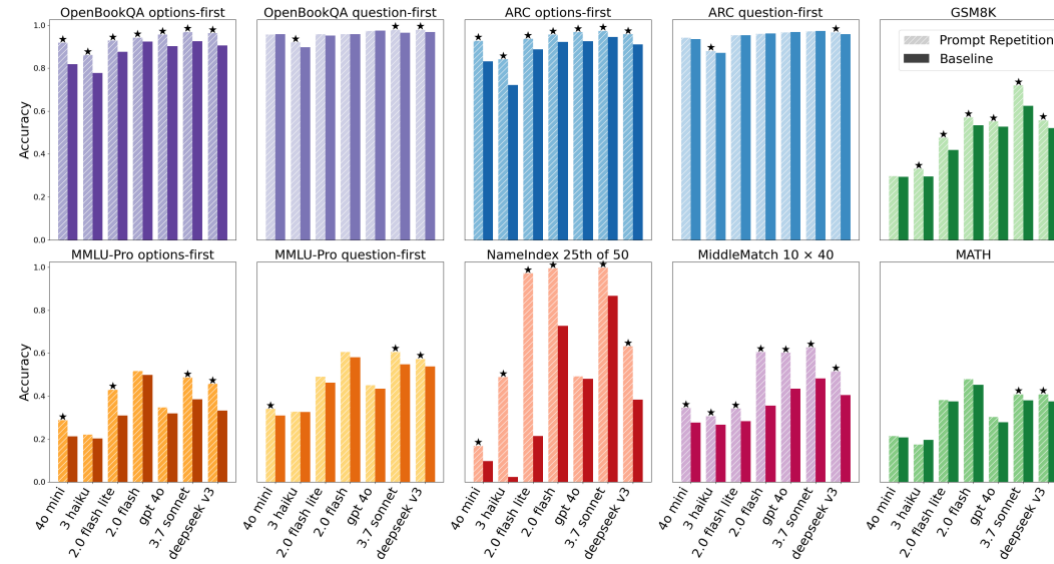
Models

- Gemini 2.0 Flash
- Gemini 2.0 Flash Lite
- GPT-4o
- GPT-4o-mini
- Claude 3 Haiku
- Claude 3.7 Sonnet
- DeepSeek V3

Benchmarks

- ARC (Challenge)
- OpenBookQA
- GSM8K
- MMLU-Pro
- MATH
- Custom tasks (NameIndex, MiddleMatch)



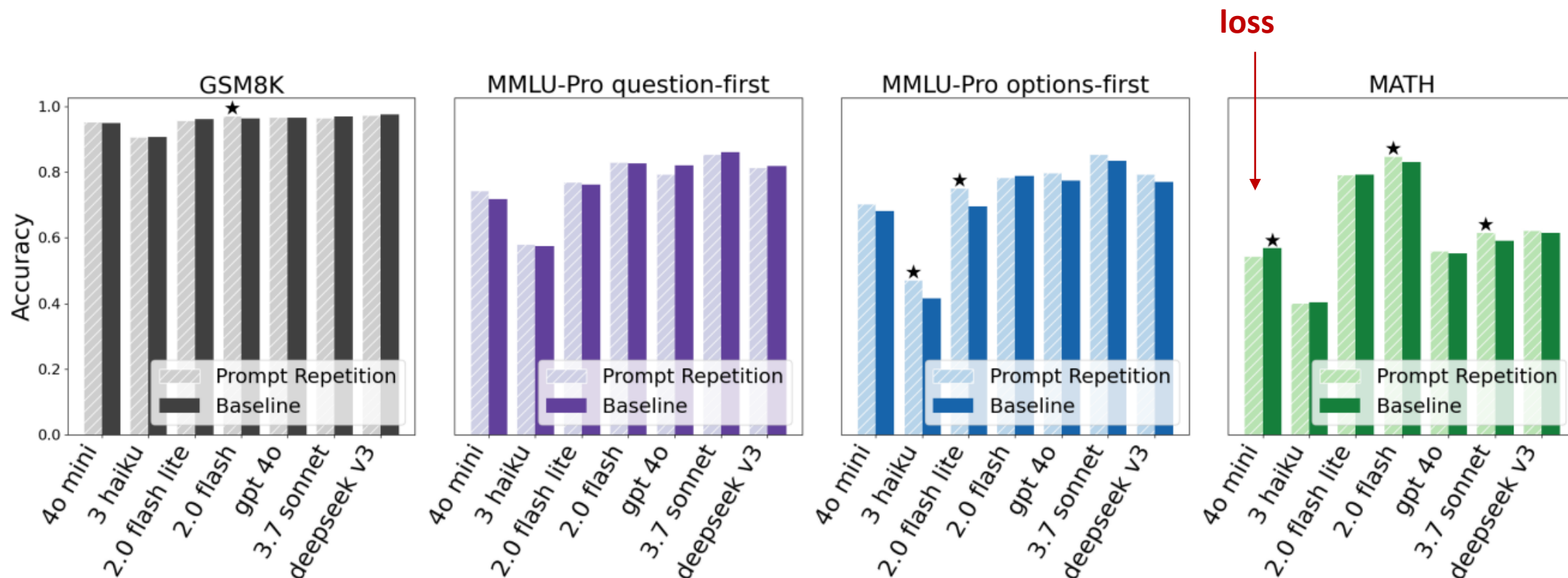


**Prompt repetition won 47 out of 70 tests,
with 0 losses**

Performance Gains in Non-Reasoning Mode

- Accuracy consistently improves.
- Output length does not increase.
- Latency remains the same.

Prompt Repetition with Reasoning



5 wins, 1 loss, 22 ties

Custom Task Results

- **NameIndex Task**

- receives 50 names and must return the 25th name.

Without repetition:

Gemini Flash Lite: 21%

With repetition:

Gemini Flash Lite: 97%

- **MiddleMatch Task:** large improvements across models.

Efficiency Analysis

- Average response length
- Median response length
- Average latency

Results:

- No increase in generation length.
- No increase in latency.

Variants Tested

- Prompt Repetition (Verbose)
- Prompt Repetition ×3
- Padding (control experiment)

Method	Template	Example Query
Prompt Repetition (Verbose)	<QUERY> Let me repeat that: <QUERY>	Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>. Let me repeat that: Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>.

Method	Template	Example Query
Prompt Repetition ×3		Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>. Let me repeat that:
	<QUERY> Let me repeat that: <QUERY> Let me repeat that one more time: <QUERY>	Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>. Let me repeat that one more time:
		Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>.

Method	Template	Example Query
Padding	<QUERY> Ignore these periods (they are irrelevant) and answer the above question:<LEN(QUERY)>....	Which of the following combinations is a mixture rather than a compound? A. oxygen and nitrogen in air B. sodium and chlorine in salt C. hydrogen and oxygen in water D. nitrogen and hydrogen in ammonia Reply with one letter ('A', 'B', 'C', 'D') in the format: The answer is <ANSWER>.. Ignore these periods (they are irrelevant) and answer the above question:

Results



Future Directions

- Fine-tune models on repeated prompts.
- Keep only second repetition in KV cache.
- Periodically repeat last generated tokens.
- Repeat only parts of prompt.
- Analyze attention maps deeply.
- Apply repetition to images.
- Use repetition with selective attention.
- Explore Prefix LM interactions.